Local cache and database traffic

- Only a request that needs this token triggers a cache check.
- Valid memory entry + successful SQL check < 1 s ago → reuse locally.
- Otherwise SELECT this token id; idle entries do not poll.
- Steady traffic: ≈ 1 token-state read / s / active id / replica.
- Pool connections do not each run a one-second query.
- Separately: 2 binding reads per successful credentialed request.
- Refresh contenders use temporary 100 ms polls; work timeout 20 s.
- Restarted replicas reuse a valid stored token on first use.
- No pub-sub / NOTIFY; peers observe changes on their next check.

Failure, invalidation and rotation

- OAuth failure → release lease; shared retry_after = now + 2 s.
- Owner crash → next contender may claim after lease expiry.
- Owner/version fence rejects a late publication by an old owner.
- An ambiguous external timeout may still cause another login.
- Admin invalidation: version++; clear token, expiry and lease.
- Peers stop reusing it after request-driven local freshness expires.
- Mounted source config change → new HMAC id; Secret updates are eventual.
- Storage-key rotation and upstream-401 refresh/replay are not implemented.
- No SQL row lock or connection is held across the provider call.